

The Ethical Hacker's Command Handbook

Chapter 2 – Part 2: Port Scanning

Learn what to scan, why to scan it, and when to use each Nmap scan type.

Learning Objectives

Understand ports, TCP vs UDP, the TCP three-way handshake, SYN, Connect and UDP scans, and learn when each scan is appropriate.

What is a Port?

A port is a logical communication endpoint. An IP address identifies the host, while the port identifies the service running on that host (for example SSH on 22 or HTTP on 80).

Why Scan Ports?

Port scanning helps identify exposed services, understand the target's attack surface, and plan the next stage of an authorized security assessment.

TCP vs UDP

TCP is connection-oriented and reliable. **UDP** is connectionless and faster but does not guarantee delivery. Most penetration testing begins with TCP because most common services use it.

TCP Three-Way Handshake

Client → SYN → Server → SYN/ACK → Client → ACK. This handshake establishes a TCP connection. Some scan types intentionally stop before the final ACK.

sudo nmap -sS

Purpose: TCP SYN (half-open) scan

What it does: Default TCP scan when you have administrator/root privileges.

When to use: Fast initial TCP enumeration in most labs.

Avoid when: When you lack administrator/root privileges.

Next Step: Next: -sV then -sC

Memory Tip: S = SYN

nmap -sT

Purpose: TCP Connect scan

What it does: Completes the full TCP handshake.

When to use: When running as a normal user without raw packet privileges.

Avoid when: When -sS is available with appropriate privileges.

Next Step: Next: -sV

Memory Tip: T = TCP Connect

sudo nmap -sU

Purpose: UDP scan

What it does: Scans UDP ports such as DNS (53), NTP (123), and SNMP (161).

When to use: After TCP scans or when UDP services are expected.

Avoid when: When only a quick TCP assessment is required.

Next Step: Next: Investigate detected UDP services.

Memory Tip: U = UDP

Comparison Table

Scan	Speed	Privileges	Typical Use
-sS	Fast	Root/Admin	Primary TCP enumeration
-sT	Moderate	None	When -sS isn't available
-sU	Slow	Root/Admin recommended	Discover UDP services

Quick Workflow

Target → **-sS** → **-sV** → **-sC** → Research findings → Document results

Revision Notes

- Port = Service endpoint
- TCP = Reliable
- UDP = Faster, connectionless
- -sS = Main TCP scan
- -sT = Use without admin/root privileges
- -sU = Scan UDP services

Practice Questions

1. Why is -sS called a half-open scan?
2. When would you choose -sT over -sS?
3. Name three common UDP services.
4. What command usually follows a successful -sS scan?